

7. Symmetric Matrices and Quadratic Forms

7.1 Diagonalization of Symmetric Matrices

Theorem 1

If A is symmetric, then any two eigenvectors from different eigenvalues are orthogonal.

Definition: Orthogonally Diagonalizable

A $n \times n$ matrix A is orthogonally diagonalizable if there exists an orthogonal matrix P and a diagonal matrix D such that $A = PDP^{-1} = PDP^T$ (since P is orthogonal, $P^{-1} = P^T$).

Theorem 2

A $n \times n$ matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

Theorem 3: The Spectral Theorem for Symmetric Matrices

An $n \times n$ symmetric matrix A has the following properties:

- A has n real eigenvalues, counting multiplicities.
- The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.
- The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.
- A is orthogonally diagonalizable.

How to Find P ?

- Find the eigenvectors of A .
- Eigenvectors corresponding to different eigenvalues are orthogonal (by Theorem 3c).
- Normalize all eigenvectors to get orthonormal eigenvectors $\vec{u}_1, \dots, \vec{u}_n$.

Spectral Decomposition of a Matrix

Suppose $A = PDP^{-1}$, where the columns of P are orthonormal eigenvectors $\vec{u}_1, \dots, \vec{u}_n$ of A and the corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ are in the diagonal matrix D . Then, since $P^{-1} = P^\top$,

$$\begin{aligned} A = PDP^\top &= (\vec{u}_1 \dots \vec{u}_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{pmatrix} \\ &= (\lambda_1 \vec{u}_1 \dots \lambda_n \vec{u}_n) \begin{pmatrix} \vec{u}_1^\top \\ \vdots \\ \vec{u}_n^\top \end{pmatrix} \\ &= \lambda_1 \vec{u}_1 \vec{u}_1^\top + \dots + \lambda_n \vec{u}_n \vec{u}_n^\top \end{aligned}$$

This is called a spectral decomposition of A .

7.2 Quadratic Forms

Definition: Quadratic Forms

A quadratic form on $\mathbb{R}^n$ is a function Q defined on $\mathbb{R}^n$ whose value at a vector $\vec{x}$ in $\mathbb{R}^n$ is computed by $Q(\vec{x}) = \vec{x}^\top A \vec{x}$, where A is an $n \times n$ symmetric matrix (called the matrix of the quadratic form).

Change of Variable in a Quadratic Form

If $\vec{x}$ represents a vector in $\mathbb{R}^n$, then a change of variable is an equation of the form $\vec{x} = P\vec{y}$ (or $\vec{y} = P^{-1}\vec{x}$), where P is an invertible matrix and $\vec{y}$ is a new "variable vector". The columns of P determine the new coordinate system.

For a quadratic form $Q(\vec{x}) = \vec{x}^\top A \vec{x}$, substituting $\vec{x} = P\vec{y}$ gives:

$$\vec{x}^\top A \vec{x} = (P\vec{y})^\top A P \vec{y} = \vec{y}^\top P^\top A P \vec{y} = \vec{y}^\top (P^\top A P) \vec{y}$$

The new matrix of the quadratic form is $P^\top A P$. Since A is symmetric, Theorem 2 guarantees that there exists an orthogonal matrix P such that $P^\top A P$ is a diagonal matrix D , and the new quadratic form is $\vec{y}^\top D \vec{y}$ (with no cross-product terms).

Theorem 4: The Principal Axes Theorem

Let A be an $n \times n$ symmetric matrix. Then there is an orthogonal change of variable $\vec{x} = P\vec{y}$ that transforms the quadratic form $\vec{x}^\top A \vec{x}$ into a quadratic form $\vec{y}^\top D \vec{y}$ with no cross-product terms.

Definition: Definiteness of Quadratic Forms

A quadratic form Q is:

- positive semidefinite** if $Q(\vec{x}) \geq 0$ for all $\vec{x} \neq \vec{0}$;
- negative definite** if $Q(\vec{x}) < 0$ for all $\vec{x} \neq \vec{0}$;
- negative semidefinite** if $Q(\vec{x}) \leq 0$ for all $\vec{x} \neq \vec{0}$;
- indefinite** if $Q(\vec{x})$ assumes both positive and negative values.

Theorem 5: Quadratic Forms and Eigenvalues

Let A be an $n \times n$ symmetric matrix. Then the quadratic form $\vec{x}^\top A \vec{x}$ is:

- positive definite if and only if all eigenvalues of A are positive;
- negative definite if and only if all eigenvalues of A are negative;
- indefinite if and only if A has both positive and negative eigenvalues.

Definitions of Definite Matrices

- A **positive definite matrix** A is a symmetric matrix for which the quadratic form $\vec{x}^\top A \vec{x}$ is positive definite.
- A **positive semidefinite matrix** is defined analogously, as are negative definite and negative semidefinite matrices.

Supplementary Properties of Symmetric Positive Definite Matrices

- If S and T are symmetric positive definite matrices, then $S + T$ is also a symmetric positive definite matrix.
- A symmetric matrix S is positive definite if and only if it satisfies any (and hence all) of the following properties:
 - All leading principal minors of S are positive;
 - All pivots of S are positive;
 - All n eigenvalues of S are positive;
 - $\vec{x}^\top S \vec{x} > 0$ for all $\vec{x} \neq \vec{0}$;
 - $S = A^\top A$ where the columns of A are linearly independent.

Inertia Theorem

Let the rank of the quadratic form $f = \vec{x}^\top A \vec{x}$ be r . If there exist two invertible transformations $\vec{x} = C\vec{y}$ and $\vec{x} = P\vec{z}$ such that:

$$f = k_1 y_1^2 + k_2 y_2^2 + \cdots + k_r y_r^2 \quad (k_i \neq 0)$$

and

$$f = \lambda_1 z_1^2 + \lambda_2 z_2^2 + \cdots + \lambda_r z_r^2 \quad (\lambda_i \neq 0)$$

then the number of positive values among k_i is equal to the number of positive values among λ_i , and the number of negative values among k_i is equal to the number of negative values among λ_i .

- The number of positive values is called the **positive inertia index**;
- The number of negative values is called the **negative inertia index**.

7.3 Constrained Optimization

Theorem 6

Let A be a symmetric matrix, and define $m = \min\{\vec{x}^\top A \vec{x} \mid \vec{x}^\top \vec{x} = 1\}$ and $M = \max\{\vec{x}^\top A \vec{x} \mid \vec{x}^\top \vec{x} = 1\}$. Then:

- M is the greatest eigenvalue of A , achieved when $\vec{x}$ is a unit eigenvector corresponding to M ;
- m is the least eigenvalue of A , achieved when $\vec{x}$ is a unit eigenvector corresponding to m .

Theorem 7

Let A be a symmetric matrix, and let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ be its eigenvalues. Then the minimum value of $\vec{x}^\top A \vec{x}$ subject to the constraints $\vec{x}^\top \vec{x} = 1$ and $\vec{x}^\top \vec{u}_1 = 0$ (where $\vec{u}_1$ is a unit eigenvector corresponding to λ_1) is λ_2 , achieved when $\vec{x}$ is a unit eigenvector $\vec{u}_2$ corresponding to λ_2 .

Theorem 8

Let A be an $n \times n$ symmetric matrix with orthogonal diagonalization $A = PDP^\top$, where the diagonal entries of D are arranged as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and the columns of P are the corresponding unit eigenvectors $\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_n$. Then for each $k = 1, 2, \cdots, n$, the maximum value of $\vec{x}^\top A \vec{x}$ subject to the constraints:

$$\vec{x}^\top \vec{x} = 1, \quad \vec{x}^\top \vec{u}_1 = 0, \quad \dots, \quad \vec{x}^\top \vec{u}_{k-1} = 0$$

is λ_k , achieved when $\vec{x} = \vec{u}_k$.

7.4 Singular Value Decomposition (SVD)

Definition: Singular Values

The **singular values** of an $m \times n$ matrix A are the square roots of the eigenvalues of $A^\top A$, denoted by $\sigma_1, \sigma_2, \dots, \sigma_n$, and arranged in decreasing order: $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, 2, \dots, n$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ are the eigenvalues of $A^\top A$.

Singular Values as Lengths of $A\vec{v}_i$

The singular values of A are the lengths of the vectors $A\vec{v}_1 \dots A\vec{v}_n$.

Proof

Suppose the eigenvectors of $A^\top A$ are $\vec{v}_1 \dots \vec{v}_n$.

First, compute the squared norm of $A\vec{v}_i$:

$$\|A\vec{v}_i\|^2 = (A\vec{v}_i)^\top A\vec{v}_i = \vec{v}_i^\top A^\top A\vec{v}_i$$

Since $\vec{v}_i$ is an eigenvector of $A^\top A$, $A^\top A\vec{v}_i = \lambda_i \vec{v}_i$, so:

$$\|A\vec{v}_i\|^2 = \vec{v}_i^\top \lambda_i \vec{v}_i = \lambda_i \vec{v}_i^\top \vec{v}_i$$

Then suppose $\{\vec{v}_1 \dots \vec{v}_n\}$ is an **orthonormal basis** (so $\vec{v}_i^\top \vec{v}_i = 1$), thus:

$$\|A\vec{v}_i\|^2 = \lambda_i$$

Note (P.S.)

$\vec{v}_i$ 为特征向量构成的标准正交基。

Theorem 9

Suppose $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $A^\top A$, arranged so that the corresponding eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Suppose A has r

nonzero singular values (i.e., $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ and $\sigma_{r+1} = \dots = \sigma_n = 0$). Then:

- $\{A\vec{v}_1, A\vec{v}_2, \dots, A\vec{v}_r\}$ is an orthogonal basis for $\text{Col } A$;
- $\text{rank } A = r$.

Definition: Σ

The decomposition of A involves an $m \times n$ "diagonal" matrix Σ :

$$\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$$

where D is an $r \times r$ diagonal matrix ($r \leq \min\{m, n\}$)

Theorem 10: The Singular Value Decomposition (SVD)

Let A be an $m \times n$ matrix with rank r . Then there exists:

- An $m \times m$ orthogonal matrix U (left singular vectors);
- An $n \times n$ orthogonal matrix V (right singular vectors);
- A diagonal matrix Σ of size $m \times n$, where the diagonal entries are the first r singular values of A : $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, and the remaining entries are 0;

such that:

$$A = U\Sigma V^\top$$

PROOF

Let λ_i and $\vec{v}_i$ be as in Theorem 9, so that $\{A\vec{v}_1, \dots, A\vec{v}_r\}$ is an orthogonal basis for $\text{Col } A$.

Normalize each $A\vec{v}_i$ to obtain an orthonormal basis $\{\vec{u}_1 \dots \vec{u}_r\}$:

$$\vec{u}_i = \frac{1}{\|A\vec{v}_i\|} A\vec{v}_i = \frac{1}{\sigma_i} A\vec{v}_i$$

and

$$A\vec{v}_i = \sigma_i \vec{u}_i \quad (1 \leq i \leq r)$$

Now extend $\{\vec{u}_1 \dots \vec{u}_r\}$ to an orthonormal basis $\{\vec{u}_1 \dots \vec{u}_m\}$ of $\mathbb{R}^m$, and let:

$$U = (\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m), \quad V = (\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$$

And then:

$$\begin{aligned}
 AV &= [A\vec{v}_1 \ \dots \ A\vec{v}_r \ 0 \ \dots \ 0] = [\sigma_1\vec{u}_1 \ \dots \ \sigma_r\vec{u}_r \ 0 \ \dots \ 0] \\
 U\Sigma &= [\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_r \ \dots \ \vec{u}_m] \begin{bmatrix} \sigma_1 & & & & 0 \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ & & & & 0 \end{bmatrix} \\
 &= (\sigma_1\vec{u}_1 \ \sigma_2\vec{u}_2 \ \dots \ \sigma_r\vec{u}_r \ 0 \ \dots \ 0)
 \end{aligned}$$

So

$$AV = U\Sigma$$

And since V is an orthogonal matrix:

$$A = U\Sigma V^\top$$

Theorem: The Invertible Matrix Theorem (concluded)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix:

- u. $(\text{Col } A)^\perp = \{0\}$
- v. $(\text{Nul } A)^\perp = \mathbb{R}^n$
- w. $\text{Row } A = \mathbb{R}^n$
- x. A has n nonzero singular values.

Theorem 11: Orthogonal Bases from SVD

Given an SVD $A = U\Sigma V^\top$ of an $m \times n$ matrix A , let $r = \text{rank } A$. Then:

- $\{\vec{u}_1, \dots, \vec{u}_r\}$ is an orthonormal basis for $\text{Col } A$;
- $\{\vec{u}_{r+1}, \dots, \vec{u}_m\}$ is an orthonormal basis for $\text{Null } A^\top$ (left null space);
- $\{\vec{v}_1, \dots, \vec{v}_r\}$ is an orthonormal basis for $\text{Row } A$;
- $\{\vec{v}_{r+1}, \dots, \vec{v}_n\}$ is an orthonormal basis for $\text{Null } A$ (null space).

Reduced Singular Value Decomposition

Let $U_r = [\vec{u}_1 \ \dots \ \vec{u}_r]$ ($m \times r$), $V_r = [\vec{v}_1 \ \dots \ \vec{v}_r]$ ($n \times r$), and $D = \text{diag}(\sigma_1, \dots, \sigma_r)$ ($r \times r$ diagonal matrix). Then the reduced SVD of A is:

$$A = U_r D V_r^\top$$

Moore-Penrose Inverse (Pseudoinverse)

The Moore-Penrose inverse of A , denoted A^+ , is defined as:

$$A^+ = V_r D^{-1} U_r^\top$$

where $D^{-1} = \text{diag}(1/\sigma_1, \dots, 1/\sigma_r)$.